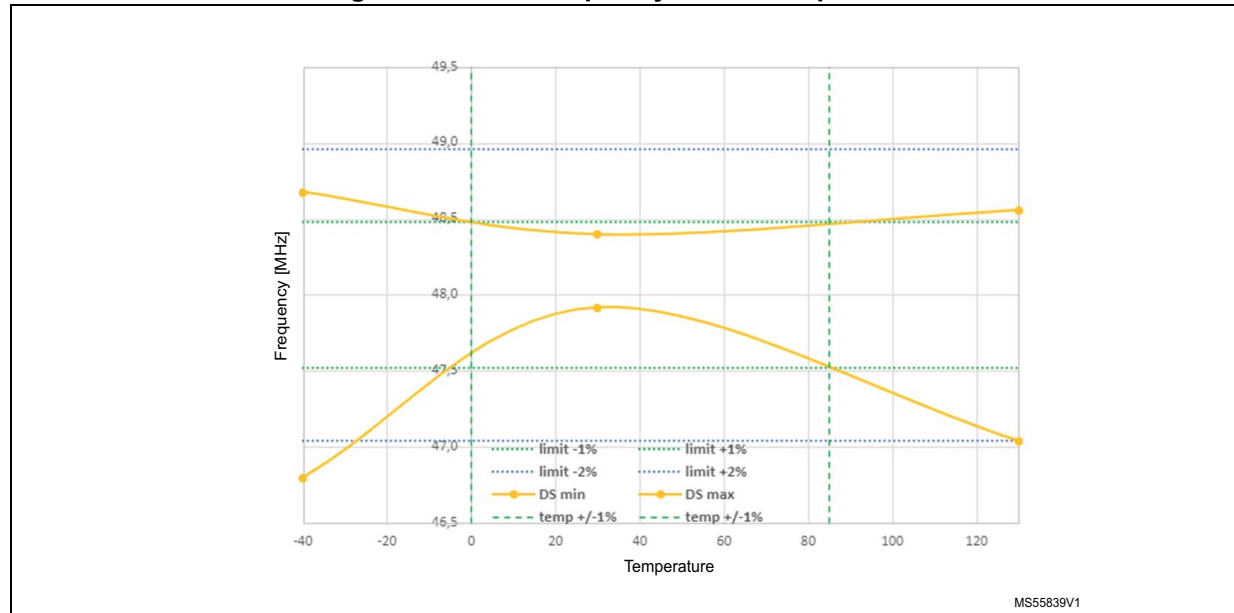


1. Based on characterization results, not tested in production
2. Specified by design. Not tested in production.

Figure 19. HSI48 frequency versus temperature



Low-speed internal (LSI) RC oscillator

Table 42. LSI oscillator characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{LSI}	LSI frequency	$V_{DD} = 3.3\text{ V}$, $T_A = 25\text{ °C}$	31.04	32	32.96	kHz
		$V_{DD} = 2\text{ V to }3.6\text{ V}$, $T_A = -40\text{ to }125\text{ °C}$	29.5 ⁽¹⁾	-	34 ⁽¹⁾	
$t_{SU(LSI)}^{(2)}$	LSI oscillator start-up time	-	-	80	130	μs
$t_{STAB(LSI)}^{(2)}$	LSI oscillator stabilization time	5% of final frequency	-	125	180	μs
$I_{DD(LSI)}^{(2)}$	LSI oscillator power consumption	-	-	110	180	nA

1. Evaluated by characterization. Not tested in production.
2. Specified by design. Not tested in production.

5.3.9 Flash memory characteristics

Table 43. Flash memory characteristics

Symbol	Parameter ⁽¹⁾	Conditions	Min	Typ	Max	Unit
t_{prog}	Word programming time	64 bits	-	85.0	125.0	μs
t_{prog_row}	Row (32 double word) programming time	Normal programming	-	2.7	4.6	ms
		Fast programming	-	1.7	2.8	